Uncertainty-Aware AI Decision Agents for Real-Time Marine and Energy Operations

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1. Strategic Alignment & Objective

As Wärtsilä advances its Veturi ecosystem toward optimised lifecycle value and data-driven operations, the demand for highly reliable, scalable artificial intelligence solutions is paramount. Operating complex hybrid marine and energy environments requires operators to make real-time decisions based on massive telemetry streams. I propose a collaborative initiative to develop **Uncertainty-Aware AI Decision Agents**—leveraging advanced methodologies in metacognitive architectures and probabilistic machine learning to provide remote operators with trustworthy, explainable, and context-aware decision support.

2. The Operational Challenge

While traditional AI and machine learning models excel at predictive insights, they often function as "black boxes," providing recommendations without quantifying the statistical confidence of those outputs. In critical marine and grid-balancing environments, this lack of transparency can lead to alert fatigue, false positives, and diminished operator trust. Wärtsilä's explicit focus on "AI agents & real-time decision support" requires systems that not only identify anomalies but also communicate their reliability transparently to the human-in-the-loop.

3. The Methodological Solution

This proposal bridges the University of Turku's deep expertise in advanced AI modelling with Wärtsilä's applied development needs. By translating methodologies from learning analytics into industrial operational data, we can build robust support pipelines:

- **Confidence-Calibrated Predictions via Split Conformal Prediction:** Instead of outputting single-point predictions for system efficiency or predictive maintenance, the AI agent will generate mathematically backed confidence intervals. This guarantees that every alert or recommendation provided to a Wärtsilä operator includes a precise reliability metric, significantly reducing false positives and operational downtime.
- **Explainable Operations via Retrieval-Augmented Generation (RAG):** When an anomaly is detected, the system will utilise RAG architectures to instantly retrieve and synthesise relevant technical documentation, historical maintenance logs, and sensor telemetry. This provides the operator with immediate, actionable troubleshooting steps grounded in Wärtsilä's proprietary knowledge base.
- **Adaptive Operator Assistance:** Utilising multi-model ensemble systems and dynamic prompt routing, the AI agent can adapt its interaction style to the specific situational load of the remote operator, ensuring critical information is prioritised during high-stress interventions.

4. Proposed Pilot Initiative

To ensure the collaboration remains focused, scalable, and mutually beneficial, I propose a controlled, non-production pilot utilising historical operational data.

- **Phase 1: Ideation & Data Mapping:** Jointly identify a specific subsystem within Wärtsilä's portfolio (e.g., engine telemetry or hybrid grid balancing) where uncertainty-aware models could optimise Total Cost of Ownership (TCO).
- **Phase 2: High-Performance Prototyping:** Leverage CSC's supercomputing infrastructure (such as the LUMI supercomputer) and scalable inference APIs to train and test the multi-model ensemble on the selected historical dataset.
- **Phase 3: Validation:** Evaluate the agent's ability to provide accurate, explainable support in a simulated environment, measuring its impact on decision speed and accuracy.

5. Value Creation for Wärtsilä

By integrating conformal prediction and dynamic RAG architectures into your digital solutions, Wärtsilä can deploy AI systems that are not just predictive, but inherently trustworthy. This collaboration supports the transition from traditional technology to holistic lifecycle value by empowering operators with the insights they need to maximise reliability and performance.